

# Formalizing the Homotopy Extension Property

Mara Silge

16th July 2026

# Table of Contents

**1** Motivation

**2** Mathematics of the Homotopy Extension Property

**3** Formalization in Lean

# Table of Contents

## 1 Motivation

## 2 Mathematics of the Homotopy Extension Property

## 3 Formalization in Lean

# Goals of the thesis

- defining the homotopy extension property in Lean
- tools to prove the homotopy extension property
  - retraction criterion
  - homeomorphic space pairs inherit the homotopy extension property from one another
- proving the homotopy extension property for relative CW-complexes

# Motivation

Theorem (Cellular Approximation Theorem, [Hat02, Theorem 4.8])

*Every map  $f: X \rightarrow Y$  of CW complexes is homotopic to a cellular map. If  $f$  is already cellular on a subcomplex  $A \subset X$ , the homotopy may be taken to be stationary on  $A$ .*

Theorem (Whitehead's Theorem, [Hat02, Theorem 4.5])

*If a map  $f: X \rightarrow Y$  between connected CW complexes induces isomorphisms  $f_*: \pi_n(X) \rightarrow \pi_n(Y)$  for all  $n$ , then  $f$  is a homotopy equivalence.*

# Table of Contents

1 Motivation

2 Mathematics of the Homotopy Extension Property

3 Formalization in Lean

# The Homotopy Extension Property

Remark: All maps of topological spaces are assumed to be continuous.

## Definition (Homotopy Extension Property (HEP) [Boa16])

A pair of topological spaces  $(X, A)$  with  $A \subseteq X$  has the *homotopy extension property (HEP)* if for every topological space  $Y$ , every map  $f: X \rightarrow Y$ , and every homotopy  $H: A \times [0, 1] \rightarrow Y$  satisfying  $H(a, 0) = f(a)$  for all  $a \in A$ , there exists a homotopy  $H': X \times [0, 1] \rightarrow Y$  such that

$$\begin{aligned} H'(x, 0) &= f(x) && \text{for all } x \in X, \\ H'(a, t) &= H(a, t) && \text{for all } (a, t) \in A \times [0, 1]. \end{aligned}$$

- $(X, X)$  has the HEP
- $(X, \emptyset)$  has the HEP

# Retraction

Let  $(X, A)$  be a pair of topological spaces with  $A \subseteq X$ .

## Definition (Retraction)

A map  $r: X \rightarrow A$  is a *retraction* if  $r \circ \text{incl}_A = \text{id}_A$ .

Equivalently, the function  $r: X \rightarrow A$

- is continuous,
- is fixed on  $A$ , i.e.  $r(a) = a$  for all  $a \in A$ .



# Retraction Criterion

## Lemma (Retraction Criterion)

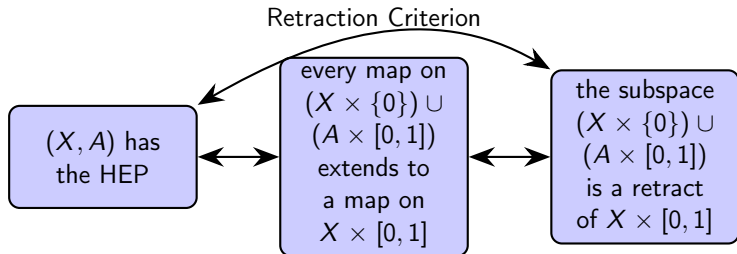
Let  $X$  be a topological space,  $A \subseteq X$  a subspace. Consider the following two statements:

- 1 The pair  $(X, A)$  has the HEP.
- 2 The subspace  $(X \times \{0\}) \cup (A \times [0, 1])$  is a retract of  $X \times [0, 1]$ .

Then  $(1) \implies (2)$ . If, in addition,  $A$  is closed in  $X$ , the converse  $(2) \implies (1)$  holds as well.

# Retraction Criterion

Let  $X$  be a topological space and  $A \subseteq X$  a closed subspace.



# HEP for relative CW-complexes

Let  $X$  be a CW-complex with skeleta  $(X_n)_{n \in \mathbb{N}}$ .

Strategy:

- 1 Any disc relative to its boundary has the *HEP*.
- 2 The pair of two consecutive skeleta has the *HEP*, i.e.  $(X_{n+1}, X_n)$  has the *HEP* for all  $n$ .
- 3 Any relative CW-complex has the *HEP*.

# HEP for discs/cubes relative boundary

Theorem ([Boa16])

*The pair  $(D^m, S^{m-1})$  has the HEP for all  $m \in \mathbb{N}$ .*

# HEP for discs/cubes relative boundary

## Theorem ([Boa16])

*The pair  $(D^m, S^{m-1})$  has the HEP for all  $m \in \mathbb{N}$ .*

## Theorem

*The HEP is preserved under homeomorphisms: let  $(X, A)$  and  $(Y, B)$  be space pairs. If there exists a homeomorphism from  $X$  to  $Y$  that maps  $A$  to  $B$ , and  $(X, A)$  has the HEP, then so does  $(Y, B)$ .*

# HEP for discs/cubes relative boundary

## Theorem ([Boa16])

*The pair  $(D^m, S^{m-1})$  has the HEP for all  $m \in \mathbb{N}$ .*

## Theorem

*The HEP is preserved under homeomorphisms: let  $(X, A)$  and  $(Y, B)$  be space pairs. If there exists a homeomorphism from  $X$  to  $Y$  that maps  $A$  to  $B$ , and  $(X, A)$  has the HEP, then so does  $(Y, B)$ .*

## Theorem

*The pair  $([0, 1]^m, \partial([0, 1]^m))$  has the HEP for all  $m \in \mathbb{N}$ .*

# Table of Contents

1 Motivation

2 Mathematics of the Homotopy Extension Property

3 Formalization in Lean

# Definition of the HEP (1/2)

- `agreeOn` isolates the compatibility condition  $H(a, 0) = f(a)$
- homotopies have type  $X \times \mathbb{R} \rightarrow Y$ , not  $A \times [0, 1] \rightarrow Y$
- `HomotopyExtension` : exactly the three conditions of the desired homotopy

```

universe u
variable {X Y : Type u} [TopologicalSpace X] [TopologicalSpace Y] {A :
  Set X}

def agreeOn (f : X → Y) (H : X × ℝ → Y) (A : Set X) : Prop := ∀ (a :
  A), f a = H (a, 0)

def HomotopyExtension (H' : X × ℝ → Y) (f : X → Y) (H : X × ℝ → Y)
  (A : Set X) :
  Prop :=
ContinuousOn H' { p : X × ℝ | p.2 ∈ unitInterval } ∧
(∀ (x : X), f x = H' (x, 0)) ∧ (∀ (a : A) (t : unitInterval), H (a, t)
  = H' (a, t))

```



## Definition of the HEP (2/2)

```
def HEP (X : Type u) [TopologicalSpace X] (A : Set X) : Prop :=  
  ∀ (Y : Type u) [TopologicalSpace Y], ∀ (f : X → Y),  
  ∀ (H : X × ℝ → Y), Continuous f →  
  ContinuousOn H {p : X × ℝ | (p.1 ∈ A) ∧ (p.2 ∈ (unitInterval))} →  
  agreeOn f H A → ∃ (H' : X × ℝ → Y), HomotopyExtension H' f H A
```

# HEP for a subset instead of a subtype

- motivation: in a CW-complex all skeleta  $X_n$  are *subsets* of one ambient space
- HEP\_HEP': the two formulations really are equivalent, so nothing is lost

open `Set`.Notation

```
def HEP' {Y : Type*} [TopologicalSpace Y] (X A : Set Y) : Prop :=
  HEP X (X ↓∩ A)
```

```
lemma HEP_HEP' {Y : Type*} [TopologicalSpace Y] (X : Set Y)
  (A : Set (Set.Elem X)) :
  HEP (X.Elem) A ↔ HEP' X (Subtype.val '' A) := by
  unfold HEP'
  rw [Set.preimage_val_image_val_eq_self]
```

# Retraction Criterion in Lean

```

structure RetractionOn {X : Type*} [TopologicalSpace X]
  (r : X → X) (B A : Set X) : Prop where
  subset : A ⊆ B
  continuousOn : ContinuousOn r B
  mapsTo : B.MapsTo r A
  fixesOn : ∀ a ∈ A, r a = a

```

```

lemma retraction_criterion_closed (hA1 : IsClosed A) :
  HEP X A ↔ ∃ r : X × ℝ → X × ℝ,
  RetractionOn r {p : X × ℝ | p.2 ∈ unitInterval}
  {p : X × ℝ | p.2 = 0 ∨ (p.1 ∈ A) ∧ p.2 ∈ unitInterval}

```

```

lemma retraction_criterion_closed' {Y : Type u} [TopologicalSpace Y] (A
  X : Set Y) (hAX : A ⊆ X)
  (hA1 : IsClosed A) : HEP' X A ↔ ∃ r : Y × ℝ → Y × ℝ,
  RetractionOn r
  {p : Y × ℝ | p.1 ∈ X ∧ p.2 ∈ unitInterval}
  {p : Y × ℝ | p.1 ∈ X ∧ p.2 = 0 ∨ (p.1 ∈ A) ∧ p.2 ∈ unitInterval}

```

# Transport along a partial homeomorphism

- global homeomorphism  $X \simeq Y$  would be stronger
- HEP' is more useful for partial homeomorphism
- subspaces have to be closed to apply the retraction criterion

```
lemma PartialHomeomorph_HEP' {X Y : Type*} [TopologicalSpace X]
  [TopologicalSpace Y] {X1 X2 : Set X} (hX : X2 ⊆ X1) {Y1 Y2 : Set Y}
  (hY : Y2 ⊆ Y1) (hHEP : HEP' X1 X2) (f : PartialHomeomorph X Y)
  (hf_source : f.source = X1) (hf_target : f.target = Y1)
  (h2 : f '' X2 = Y2) (hX2closed : IsClosed (X2 : Set X))
  (hY2closed : IsClosed (Y2 : Set Y)) : HEP' Y1 Y2
```

# HEP for the disc relative to its boundary

- use  $H'$ : explicitly constructed earlier
- the primed version follows by exact: both statements are definitionally equal

```
lemma HEP_disc_boundary : ∀ (m : ℕ),
  HEP (Metric.closedBall (0 : EuclideanSpace ℝ (Fin m)) 1) (Metric.
    sphere ⟨0, by simp⟩ 1) := by
  intro m Y hY f H hf_cont hH_cont hAgree
  use H' f H
  refine ⟨?_ , ?_ , ?_ ⟩
  · exact H'_ContinuousOn f H hf_cont hH_cont hAgree
  · intro x
    simp [H', aux_fun, mem_closedBall_zero_iff.mp x.prop]
  · exact H'_agreeH f H hAgree
```

```
lemma HEP_disc_boundary' : ∀ (m : ℕ),
  HEP' (Metric.closedBall (0 : EuclideanSpace ℝ (Fin m)) 1) (Metric.
    sphere 0 1) := by
  exact HEP_disc_boundary
```

# From the disc to the cube

```

theorem exists_homeomorph_image_interior_closure_frontier_eq_unitBall      source
  {E : Type u_1} [NormedAddCommGroup E] [NormedSpace ℝ E] {s : Set E}
  (hc : Convex ℝ s) (hne : (interior s).Nonempty) (hb : Bornology.IsBounded s)
  :
  ∃ (h : E ≃ₜ E),
    ∃ h '' interior s = Metric.ball 0 1 ∧ ∃ h '' closure s =
      Metric.closedBall 0 1 ∧ ∃ h '' frontier s = Metric.sphere 0 1

```

If  $s$  is a convex bounded set with a nonempty interior in a real normed space, then there is a homeomorphism of the ambient space to itself that sends the interior of  $s$  to the unit open ball and the closure of  $s$  to the unit closed ball.

- EuclideanSpace carries the  $\ell^2$ -norm, the type  $\text{Fin } m \rightarrow \mathbb{R}$  the  $\ell^\infty$ -norm
- the above lemma matches closure with closure and frontier with frontier

```
def fun_euclid_max : EuclideanSpace ℝ (Fin m) → (Fin m → ℝ) := fun p ↦ p
```

```

def G : (Fin m → ℝ) ≃ₜ (Fin m → ℝ) := (
  exists_homeomorph_image_interior_closure_frontier_eq_unitBall
  convex_euclid_to_max_ball
  nonempty_euclid_to_max_ball bounded_euclid_to_max_ball).choose

```



# References I

- [Boa16] J. Michael Boardman. *The Homotopy Extension Property*. 2016.  
URL: <https://math.jhu.edu/~jmb/note/hep.pdf> (visited on 07/13/2026).
- [Hat02] Allen Hatcher. *Algebraic topology*. Cambridge: Cambridge University Press, 2002, pp. xii+544. ISBN: 0-521-79540-0.

Formalisation and slides: <https://github.com/marasilge/Bachelorarbeit>